

**SULIT**



## **SECP2523 DATABASE**

**SESSION 2024/2025 - SEMESTER 1**

### **ALTERNATIVE ASSESSMENT REPORT: PHASE 1**

|                            |                                  |
|----------------------------|----------------------------------|
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| <b>Year / Program</b>      | 2/SECPH                          |
| <b>Section</b>             | 2                                |
| <b>Lecturer Name</b>       | DR.Seah Choon Sean               |
| <b>Case Study</b>          | KADA Cooperation System Redesign |
| <b>System Name</b>         | KADA Cooperation                 |
| <b>Group Name</b>          | TECHMEDICO                       |

## **REPORT: PHASE 1**

### **SECTION A – SYSTEM’S OVERVIEW (15 MARKS)**

#### **1.1 Overall Description**

- 1) Provide an overview of the proposed system based on your module in the WBL project case study. Include an explanation of the current issues that prompted the initiation of the WBL project. Additionally, present the module distribution for your WBL group project in table form, as illustrated below

Our proposed KADA System is an online platform designed to digitize the cooperative’s operations, replacing the existing manual processes. Key use cases in our system include loan application, member registration, user authentication, savings management, and report generation for both administrators and members. These features aim to enhance operational efficiency, accessibility, and user satisfaction. Particularly, the report generation module for members enables KADA members to generate and access detailed monthly and annual financial reports. These reports provide information to the members about their financial status, including loan balances, transaction details and saving contribution. This transparency enables members to make informed financial decisions while fostering trust and accountability within KADA cooperative.

The initiation of the KADA System addresses several challenges especially in inefficiencies in manual processes. The amount of time needed to process applications, approve loans, and generate reports is increased by the current heavy reliance on paperwork by both KADA administrators and members. In addition, the lack of real-time updates makes it extremely difficult for users to remotely access their account and financial information. These challenges reduce the overall productivity of the KADA system and hinder the ability of members to make timely, informed decisions.

| Group’s Member Name   | Matric No | Module                                                             |
|-----------------------|-----------|--------------------------------------------------------------------|
| CHOH JING YI          | A23CS0296 | Report Generation (Members)                                        |
| Elijah She Yu Sheng   | A23CS0073 | Loan Application Module,<br>User Authentication Module             |
| Lee Yin Shen          | A23CS0236 | Membership Application Module,<br>Member Profile Management Module |
| Joanne Ching Yin Xuan | A23CS0227 | Administration Module                                              |
| Lim Yu Han            | A23CS0241 | Saving Account Module                                              |

(5 marks)

## 1.2 Project Weakness or Improvement

- 2) Based on your module in the WBL project, identify weaknesses or suggest improvements for the proposed system.

(5 marks)

One of the report generation module's weaknesses is that it only allows monthly and annual reports, which might not be sufficient for certain requirements requiring detailed evaluation. For example, members might want to filter by specific transactions history, loan types, or specific timeframes. The financial report should have detailed breakdowns of loan interest payments and each details contributions to savings.

Furthermore, members are not required to validate or confirm the financial reports that are generated. This lack of verification may cause members to make key decisions based on unconfirmed information, which could result in miscommunications or conflicts about the accuracy of the financial data. Members should be prompted by the system to confirm the report's content before downloading or completing it. Members are guaranteed to accept and agree with the validity of the report through a confirmation process.

In addition, financial reports lack detailed breakdowns such as loan interest payments or individual contributions to savings, which are essential for accurate financial analysis. The financial report should also include graphical representations of financial status, such as bar charts for loan repayment progress or pie charts for transaction distributions, to facilitate easier interpretation of the data.

- 3) Based on your answer in Section A (2), propose a new module to overcome the weaknesses.

(5 marks)

A new Comprehensive Financial Reporting and Analysis Module is proposed to overcome the weaknesses identified in question 2.

One of the main advantages is the ability to customize report production, which allows users to examine financial data by filtering loan details and transaction history for specific dates and times in both monthly and annual reports. This enables more accurate financial assessments by giving members the freedom to access the data that is most useful to their needs.

Another key feature is the implementation of a validation system, where members must confirm and verify the accuracy of financial reports before finalizing and downloading them. This lowers the possibility of inaccuracies in financial reports by guaranteeing that only reliable data will be used for decision-making.

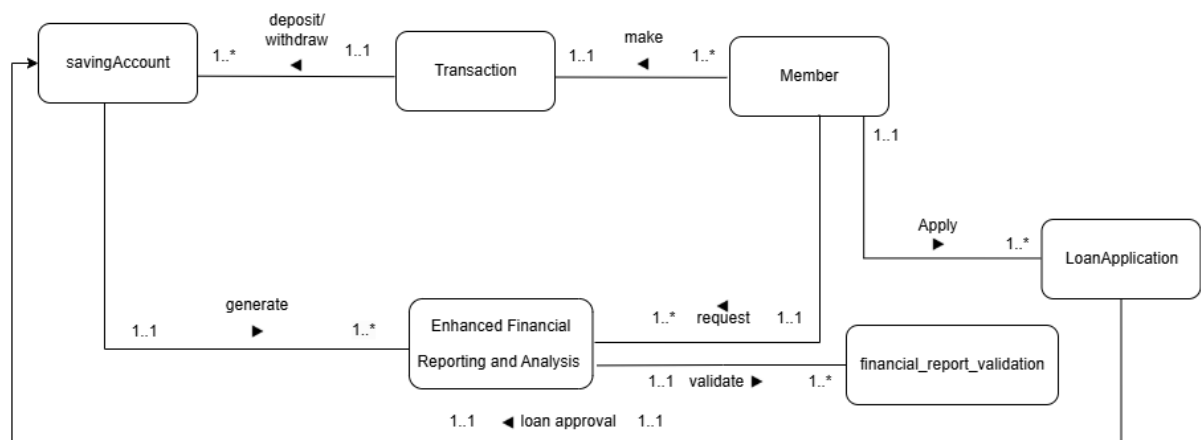
Additionally, interactive graphical tools such as bar charts, line graphs, and scatter plots can be included to help members visualize trends and patterns in their financial data effectively. Members can more easily analyse and gain deeper insights on financial performance with the use of these visual tools, which helps them make better choices.

## SECTION B – DATABASE PLANNING AND DESIGN (40 MARKS)

### 2.1 Conceptual ERD

1) Based on your answer in Section A(3), draw a **conceptual ERD** for the proposed module.

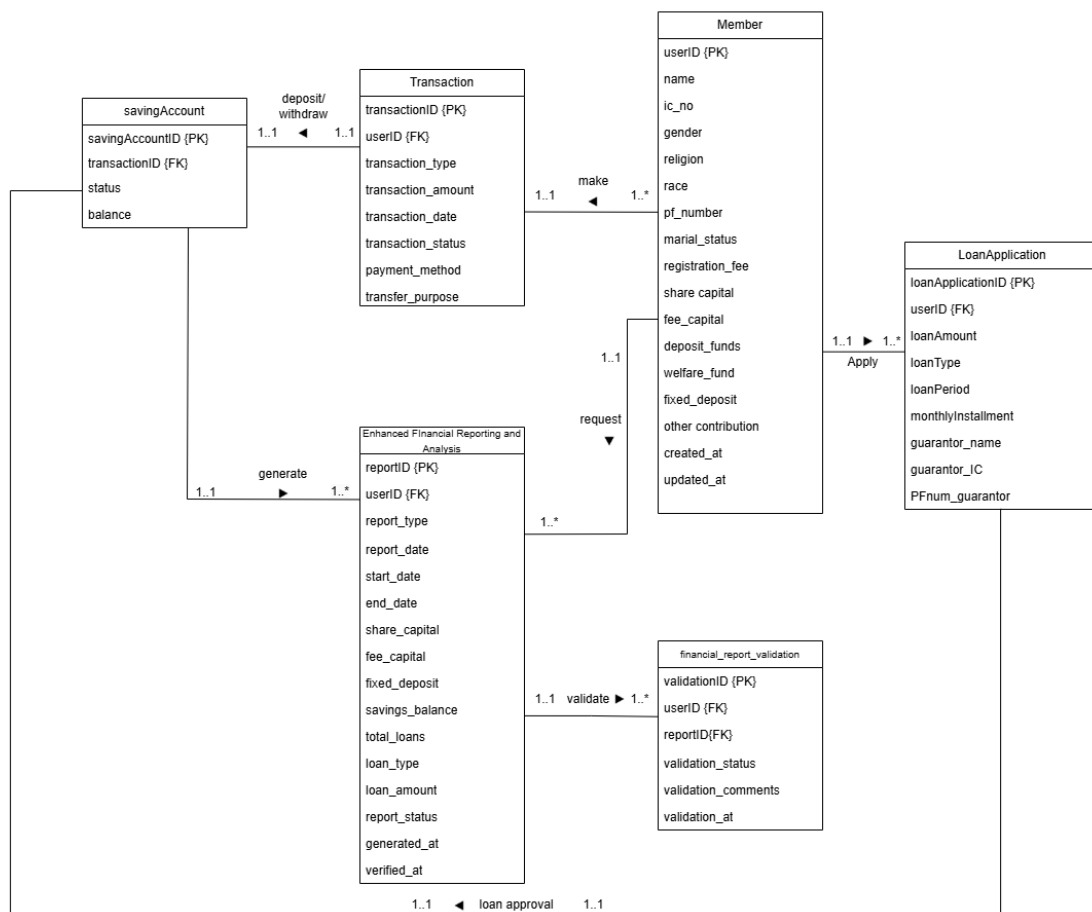
(5 marks)



### 2.2 Logical ERD

2) Create a logical ERD and data dictionary for the proposed module.

(15 marks)



## 2.3 Data Dictionary for Logical ERD

| Table Name  | Column Name        | Data type                                         | Description                                             |
|-------------|--------------------|---------------------------------------------------|---------------------------------------------------------|
| Member      | userID (PK)        | INT                                               | Unique identifier for each member.                      |
|             | name               | VARCHAR (255)                                     | Full name of the member.                                |
|             | ic_no              | VARCHAR (22)                                      | Identity card number of the member.                     |
|             | gender             | Enum('Male', 'Female')                            | Gender of the member.                                   |
|             | religion           | VARCHAR (50)                                      | Religion of the member.                                 |
|             | race               | VARCHAR (50)                                      | Race of the member.                                     |
|             | pf_number          | VARCHAR (50)                                      | PF number of the member.                                |
|             | marial_status      | Enum ('Single', 'Married', 'Divorces', 'Widowed') | Marital status of the member.                           |
|             | registration_fee   | decimal (10,2)                                    | Registration fee paid by the member.                    |
|             | share_capital      | decimal (10,2)                                    | Registration fee paid by member                         |
|             | fee_capital        | decimal (10,2)                                    | Fee capital amount                                      |
|             | deposit_funds      | decimal (10,2)                                    | Total funds deposited by member                         |
|             | welfare_fund       | decimal (10,2)                                    | Contribution to the welfare fund by the member.         |
|             | fixed_deposit      | decimal (10,2)                                    | Fixed deposit amount by the member.                     |
|             | created_at         | timestamp                                         | Date and time the record was created.                   |
|             | updated_at         | timestamp                                         | Date and time the record was last updated.              |
| Transaction | transactionID(PK)  | INT                                               | Unique identifier for the transaction.                  |
|             | userID (FK)        | INT                                               | Foreign key referencing the userID in the Member table. |
|             | transaction_amount | INT                                               | Amount involved in the transaction.                     |
|             | transaction_type   | VARCHAR (50)                                      | Type of the transaction (e.g., deposit, withdrawal)     |
|             | transaction_status | Enum('pending', 'approved', 'rejected')           | Status of transaction.                                  |
|             | transaction_date   | timestamp                                         | Date and time of the transaction was initiated          |
|             | payment_method     | VARCHAR (50)                                      | Payment method used in the transaction                  |
|             | transfer_purpose   | VARCHAR (50)                                      | Purpose or reason for the transaction.                  |

|                                           |                      |                                        |                                                                     |
|-------------------------------------------|----------------------|----------------------------------------|---------------------------------------------------------------------|
| savingAccount                             | savingAccountID (PK) | INT                                    | Unique identifier for the saving account.                           |
|                                           | transactionID (FK)   | INT                                    | Foreign key referencing the transactionID in the Transaction table. |
|                                           | balance              | decimal (10,2)                         | Amount balance in saving account                                    |
|                                           | status               | VARCHAR (20)                           | Status of the saving account (e.g., active, inactive).              |
| Enhanced Financial Reporting and Analysis | reportID(PK)         | INT                                    | Unique identifier for each financial report.                        |
|                                           | userID (FK)          | INT                                    | Foreign key referencing the userID in the Member table.             |
|                                           | report_type          | enum('monthly', 'yearly', 'custom')    | Type of report (e.g., monthly, annual, custom).                     |
|                                           | start_date           | date                                   | The start date of the report period.                                |
|                                           | end_date             | date                                   | The end date of the report period.                                  |
|                                           | share_capital        | date                                   | Represents the share capital for the member.                        |
|                                           | fee_capital          | decimal(10,2)                          | Fee capital amount                                                  |
|                                           | fixed_deposit        | decimal(10,2)                          | Total funds deposited by member.                                    |
|                                           | savings_balance      | decimal(10,2)                          | The total savings balance of the member.                            |
|                                           | total_loans          | decimal(10,2)                          | The total amount of loans                                           |
|                                           | loan_type            | VARHCAR(50)                            | The type of loan.                                                   |
|                                           | loan_amount          | decimal(10,2)                          | The amount of the loan.                                             |
|                                           | report_status        | enum('Draft', 'Verified', 'Finalized') | The status of the report.                                           |
|                                           | generated_at         | timestamp                              | The timestamp when the financial report was generated.              |
|                                           | verified_at          | timestamp                              | The timestamp when the financial report was verified.               |
| LoanApplication                           | loanAccountID(PK)    | INT                                    | Unique identifier for the loan application.                         |
|                                           | userID {FK}          | INT                                    | Foreign key referencing the userID in the Member table.             |
|                                           | loanAmount           | decimal (10,2)                         | Amount of the loan applied for.                                     |
|                                           | loanType             | VARCHAR (50)                           | Type of loan.                                                       |

|                             |                     |                                       |                                                                    |
|-----------------------------|---------------------|---------------------------------------|--------------------------------------------------------------------|
|                             |                     |                                       |                                                                    |
|                             | loanPeriod          | INT                                   | Duration of the loan in months.                                    |
|                             | monthlyInstallment  | decimal (10,2)                        | Monthly installment amount for the loan repayment.                 |
|                             | guarantor_name      | VARCHAR (50)                          | Name of the guarantor for the loan.                                |
|                             | guarantor_IC        | VARCHAR (20)                          | Identity card number of the guarantor.                             |
|                             | PFnum_guarantor     | VARCHAR (50)                          | PF number of the guarantor.                                        |
| financial_report_validation | validationID {PK}   | INT                                   | Unique identifier for the financial report validation.             |
|                             | userID {FK}         | INT                                   | Foreign key referencing the userID in the Member table.            |
|                             | reportID {FK}       | INT                                   | Foreign key referencing the reportID in the FinancialReport table. |
|                             | validation_status   | enum('Pending','Approved','Rejected') | Specifies the status of the verification.                          |
|                             | validation_comments | text                                  | This field allows to provide comments.                             |
|                             | validation_at       | timestamp                             | The timestamp of when the verification status was updated.         |

## 2.4 Relational Database

3) List all **relations** derived from the logical ERD produced in Section B(2).

10 marks

| Entity          | Relationship     | Entity2                   | Multiplicity |
|-----------------|------------------|---------------------------|--------------|
| Member          | make             | transaction               | *..1         |
| Transactions    | Deposit/withdraw | savingAccount             | 1..1         |
| SavingAccount   | generate         | financialreport           | 1..*         |
| Member          | request          | financialreport           | 1..*         |
| Member          | apply            | LoanApplication           | 1..*         |
| Member          | submit           | LoanApplication           | 1..*         |
| Financialreport | validate         | Financial_report_validate | 1..*         |

### **KADA\_database(relational database schema)**

Member(userID, name, ic\_no, gender, religion, race, pf\_number, marital\_status, registration\_fee, share\_capital, fee\_capital, deposit\_funds, welfare\_fund, fixed\_deposit, other\_contribution, created\_at, updated\_at)

PK: userID

Transaction(transactionID, userID, transaction\_type, transaction\_amount, transaction\_status, transaction\_date, transaction\_status, payment\_method, transfer\_purpose)

PK: transactionID

FK: userID

SavingAccount(savingAccountID, transactionID, balance, status)

PK: savingAccountID

FK: transactionID

Enhanced Financial Reporting and Analysis (reportID, userID, report\_type, report\_date, start\_date, end\_date, share\_capital, fee\_capital, fixed\_deposit, savings\_balance total\_loans, loan\_type, loan\_amount, report\_status, generated\_at, verified\_at)

PK: reportID

FK: userID

LoanApplication (loanAccountID, userID, loanAmount, loanType, loanPeriod, monthlyInstallment, guarantor\_name, guarantor\_IC, PFnum\_guarantor)

PK: loanAccountID

FK: userID



financial\_report\_validation(validationID, userID, reportID, validation\_status,  
validation\_comments, validation\_at)

PK: validationID

FK: userID, reportID

## 2.5 Normalization

4) Perform normalization up to **3NF** to each relation derived in Section B(3). You must show how the normalization is performed to the relations derived.

### Member relation:

**1NF:** PK: userID

| userID | name     | ic_no       | gender | religion | race    | pf_number | marital_status | registration_fee | share_capital | fee_capital | deposit_funds | welfare_fund | fixed_deposit | created_at          | updated_at          |
|--------|----------|-------------|--------|----------|---------|-----------|----------------|------------------|---------------|-------------|---------------|--------------|---------------|---------------------|---------------------|
| 1      | Ali      | 01234567890 | Male   | Buddha   | Chinese | A12       | Single         | 50               | 150           | 150         | 50            | 50           | 50            | 2025-01-11 09:48:05 | 2025-01-11 10:22:53 |
| 2      | Fati mah | 0987654321  | Female | Islam    | Malay   | A13       | Married        | 50               | 150           | 150         | 50            | 50           | 50            | 2025-01-10 18:30:14 | 2025-01-10 19:07:45 |

**2NF:**

Relation 1: Member Personal Details, PK: userID

| userID | name | ic_no | gender | religion | race | marital_status | pf_number |
|--------|------|-------|--------|----------|------|----------------|-----------|
|--------|------|-------|--------|----------|------|----------------|-----------|

|   |         |             |        |        |         |         |     |
|---|---------|-------------|--------|--------|---------|---------|-----|
| 1 | Ali     | 01234567890 | Male   | Buddha | Chinese | Single  | A12 |
| 2 | Fatimah | 0987654321  | Female | Islam  | Malay   | Married | A13 |

Relation 2: Member Financial Details, PK: userID

| userID | registration_fee | share_capital | fee_capital | deposit_funds | welfare_fund | fixed_deposit |
|--------|------------------|---------------|-------------|---------------|--------------|---------------|
| 1      | 50               | 150           | 150         | 50            | 50           | 50            |
| 2      | 50               | 150           | 150         | 50            | 50           | 50            |

Relation 3: Audit Details, PK: userID

| userID | created_at             | updated_at             |
|--------|------------------------|------------------------|
| 1      | 2025-01-11<br>09:48:05 | 2025-01-11<br>10:22:53 |
| 2      | 2025-01-10<br>18:30:14 | 2025-01-10<br>19:07:45 |

### 3NF:

Relation 1: Member Personal Details, PK: userID

| userID | name | ic_no       | gender | religion | race    | marital_status | pf_number |
|--------|------|-------------|--------|----------|---------|----------------|-----------|
| 1      | Ali  | 01234567890 | Male   | Buddha   | Chinese | Single         | A12       |

|   |         |            |        |       |       |         |     |
|---|---------|------------|--------|-------|-------|---------|-----|
| 2 | Fatimah | 0987654321 | Female | Islam | Malay | Married | A13 |
|---|---------|------------|--------|-------|-------|---------|-----|

Relation 2: Member Financial Details, PK: userID

| userID | registration_fee | share_capital | fee_capital | deposit_funds | welfare_fund | fixed_deposit |
|--------|------------------|---------------|-------------|---------------|--------------|---------------|
| 1      | 50               | 150           | 150         | 50            | 50           | 50            |
| 2      | 50               | 150           | 150         | 50            | 50           | 50            |

Relation 3: Audit Details, PK: userID

| userID | created_at             | updated_at             |
|--------|------------------------|------------------------|
| 1      | 2025-01-11<br>09:48:05 | 2025-01-11<br>10:22:53 |
| 2      | 2025-01-10<br>18:30:14 | 2025-01-10<br>19:07:45 |

## Transaction relation:

1NF: PK:transactionID

| transactionID | userID | transaction_amount | transaction_type | transaction_status | transaction_date    | payment_method | transfer_purpose |
|---------------|--------|--------------------|------------------|--------------------|---------------------|----------------|------------------|
| 1234          | 1      | 250                | deposit          | approved           | 2025-01-11 10:53:19 | debit card     | Loan repayment   |
| 1235          | 2      | 300                | transfer         | pending            | 2025-01-11 11:47:20 | online banking | pendidikan       |
| 1236          | 3      | 350                | transfer         | pending            | 2025-01-12 23:04:27 | online banking | kesihatan        |

## 2NF:

Relation 1: Transaction Details, PK:transactionID

| transactionID | userID | transaction_amount | transaction_type | transaction_status | transaction_date    |
|---------------|--------|--------------------|------------------|--------------------|---------------------|
| 1234          | 1      | 250                | deposit          | approved           | 2025-01-11 10:53:19 |
| 1235          | 2      | 300                | transfer         | pending            | 2025-01-11 11:47:20 |

Relation 2: Payment Methods, PK:transactionID

| transactionID | payment_method |
|---------------|----------------|
| 1234          | debit card     |
| 1235          | online banking |

Relation 3: Transfer Purpose, PK:transactionID

| transactionID | transfer_purpose |
|---------------|------------------|
| 1234          | Loan repayment   |
| 1235          | pendidikan       |

**3NF:**

Relation 1: Transaction Details, PK:transactionID

| transactionID | userID | transaction_amount | transaction_type | transaction_status | transaction_date       |
|---------------|--------|--------------------|------------------|--------------------|------------------------|
| 1234          | 1      | 250                | deposit          | approved           | 2025-01-11<br>10:53:19 |
| 1235          | 2      | 300                | transfer         | pending            | 2025-01-11<br>11:47:20 |

Relation 2: Payment Methods, PK:transactionID

| transactionID | payment_method |
|---------------|----------------|
| 1234          | debit card     |
| 1235          | online banking |

Relation 3: Transfer Purpose, PK:transactionID

| transactionID | transfer_purpose |
|---------------|------------------|
| 1234          | Loan repayment   |
| 1235          | pendidikan       |

## **SavingAccount relation:**

**1NF:** PK: savingAccountID

| savingAccountID | transactionID | balance | status   |
|-----------------|---------------|---------|----------|
| 1234            | 123           | 3000    | active   |
| 1235            | 124           | 4000    | inactive |

**2NF:**

Relation 1: Saving Account Transactions

| savingAccountID | transactionID |
|-----------------|---------------|
| 1234            | 123           |
| 1235            | 124           |

Relation 2: Saving Account Details, PK: savingAccountID

| savingAccountID | balance | status   |
|-----------------|---------|----------|
| 1234            | 3000    | active   |
| 1235            | 4000    | inactive |

**3NF:**

Relation 1: Saving Account Transactions

| savingAccountID | transactionID |
|-----------------|---------------|
| 1234            | 123           |
| 1235            | 124           |

Relation 2: Saving Account Details, PK: savingAccountID

| savingAccountID | balance | status   |
|-----------------|---------|----------|
| 1234            | 3000    | active   |
| 1235            | 4000    | inactive |



## Enhanced Financial Reporting and Analysis relation:

1NF:

PK: reportID

| reportID | userID | report_type | report_date | start_date | end_date   | share_capital | fee_capital | fixed_deposit | savings_balance | total_loans | loan_type           | loan_amount | report_status | generated_at        | verified_at         |
|----------|--------|-------------|-------------|------------|------------|---------------|-------------|---------------|-----------------|-------------|---------------------|-------------|---------------|---------------------|---------------------|
| 1        | 1      | monthly     | 2025-01-31  | 2025-01-01 | 2025-01-31 | 1000.00       | 50.00       | 50.00         | 5000.00         | 4000.00     | Pembiayaan_Al_Innah | 20000.00    | Verified      | 2025-01-24 22.17.49 | 2025-01-24 22.30.21 |
| 2        | 2      | yearly      | 2025-01-01  | 2025-01-01 | 2025-12-31 | 1000.00       | 50.00       | 50.00         | 5000.00         | 4000.00     | Pembiayaan_Al_Bai   | 20000.00    | Verified      | 2025-01-22 22.07.41 | 2025-01-24 22.40.21 |

2NF:

Relation 1: FinancialReport Table, PK: reportID

| reportID | userID | report type | report date | report status | generated at        | verified at         |
|----------|--------|-------------|-------------|---------------|---------------------|---------------------|
| 1        | 1      | monthly     | 2025-01-31  | Verified      | 2025-01-24 22.17.49 | 2025-01-24 22.30.21 |

|   |   |        |            |          |                        |                        |
|---|---|--------|------------|----------|------------------------|------------------------|
| 2 | 2 | yearly | 2025-01-01 | Verified | 2025-01-22<br>22.07.41 | 2025-01-24<br>22.40.21 |
|---|---|--------|------------|----------|------------------------|------------------------|

Relation 2: MemberFinancialDetails Table, PK:userID

| userID | start_date | end_date   | share_capital | fee_capital | fixed_deposit | savings_balnce |
|--------|------------|------------|---------------|-------------|---------------|----------------|
| 1      | 2025-01-01 | 2025-01-31 | 1000.00       | 50.00       | 50.00         | 5000.00        |
| 2      | 2025-01-01 | 2025-12-31 | 1000.00       | 50.00       | 50.00         | 5000.00        |

Relation 3: LoanDetails Table Composite key (total\_loans, loan\_type, loan\_amount).

| total_loans | loan_type           | loan_amount |
|-------------|---------------------|-------------|
| 40000.00    | Pembiayaan Al Innah | 20000.00    |
| 40000.00    | Pembiayaan Al Bai   | 20000.00    |

### 3NF:

Relation 1: FinancialReport Table, PK: reportID

| reportID | userID | report_type | report_date | report_status | generated_at           | verified_at            |
|----------|--------|-------------|-------------|---------------|------------------------|------------------------|
| 1        | 1      | monthly     | 2025-01-31  | Verified      | 2025-01-24<br>22.17.49 | 2025-01-24<br>22.30.21 |
| 2        | 2      | yearly      | 2025-01-01  | Verified      | 2025-01-22<br>22.07.41 | 2025-01-24<br>22.40.21 |

Relation 2: MemberFinancialDetails Table , PK:userID

| userID | start_date | end_date   | share_capital | fee_capital | fixed_deposit | savings_balnce |
|--------|------------|------------|---------------|-------------|---------------|----------------|
| 1      | 2025-01-01 | 2025-01-31 | 1000.00       | 50.00       | 50.00         | 5000.00        |
| 2      | 2025-01-01 | 2025-12-31 | 1000.00       | 50.00       | 50.00         | 5000.00        |

Relation 3: LoanDetails Table Composite key (total\_loans, loan\_type, loan\_amount).

| total_loans | loan_type           | loan_amount |
|-------------|---------------------|-------------|
| 40000.00    | Pembiayaan Al Innah | 20000.00    |
| 40000.00    | Pembiayaan Al Bai   | 20000.00    |

## LoanApplication:

1NF:

PK: loanAccountID

| loanAccountID | userID | loanAmount | loanType            | loanPeriod | monthlyInstallment | guarantor_name | guarantor_IC | PFnum_guarantor |
|---------------|--------|------------|---------------------|------------|--------------------|----------------|--------------|-----------------|
| 1             | 1      | 10000      | Pembiayaan Al Bai   | 36         | 312.78             | Jingyi         | 1234567890   | A3              |
| 2             | 2      | 20000      | Pembiayaan Al Innah | 48         | 486.67             | jingyuan       | 0987654321   | A4              |

2NF:

Relation 1: Loan Details, PK: loanAccountID

| loanAccountID | userID | loanAmount | loanType            | loanPeriod | monthlyInstallment |
|---------------|--------|------------|---------------------|------------|--------------------|
| 123           | 1      | 10000      | Pembiayaan Al Bai   | 36         | 312.78             |
| 124           | 2      | 20000      | Pembiayaan Al Innah | 48         | 486.67             |

Relation 2: Guarantor Details, PK: loanAccountID

| loanAccountID | guarantor_name | guarantor_IC | PFnum_guarantor |
|---------------|----------------|--------------|-----------------|
| 123           | Jingyi         | 1234567890   | A3              |
| 124           | jingyuan       | 0987654321   | A4              |

### 3NF:

Relation 1: Loan Details, PK: loanAccountID

| loanAccountID | userID | loanAmount | loanType               | loanPeriod |
|---------------|--------|------------|------------------------|------------|
| 123           | 1      | 10000      | Pembiayaan<br>Al Bai   | 36         |
| 124           | 2      | 20000      | Pembiayaan<br>Al Innah | 48         |

Relation 2: Guarantor Details, PK: loanAccountID

| loanAccountID | guarantor_name | guarantor_IC | PFnum_guarantor |
|---------------|----------------|--------------|-----------------|
| 123           | Jingyi         | 1234567890   | A3              |
| 124           | jingyuan       | 0987654321   | A4              |

## Financial\_report\_validation:

1NF:

**PK:** validationID,

| validationID | reportID | userID | validation_status | validation_comments                           | validated_at        |
|--------------|----------|--------|-------------------|-----------------------------------------------|---------------------|
| 1            | 1        | 1      | Approved          | Report is valid and verified for January 2025 | 2025-01-31-10:00:00 |
| 2            | 2        | 2      | Approved          | Report is valid and verified for January 2025 | 2025-01-31-10:00:00 |

2NF:

Relation 1: FinancialReportValidation ,PK: validationID

| validationID | reportID | userID |
|--------------|----------|--------|
| 1            | 1        | 1      |
| 2            | 2        | 2      |

Relation 2: ValidationDetails ,PK: reportID

| reportID | validation_status | validation_comments                           | validated_at        |
|----------|-------------------|-----------------------------------------------|---------------------|
| 1        | Approved          | Report is valid and verified for January 2025 | 2025-01-31-10:00:00 |
| 2        | Approved          | Report is valid and verified for January 2025 | 2025-01-31-10:00:00 |

### 3NF:

Relation 1: FinancialReportVerification, PK: validationID

| validationID | reportID | userID |
|--------------|----------|--------|
| 1            | 1        | 1      |
| 2            | 2        | 2      |

Relation 2: ValidationStatus Table (linked to reportID), PK: reportID

| reportID | validation_status | validated_at        |
|----------|-------------------|---------------------|
| 1        | Approved          | 2025-01-31-10:00:00 |
| 2        | Approved          | 2025-01-31-10:00:00 |

Relation 3: ValidationComments Table (linked to reportID and userID), PK: reportID

| reportID | userID | validation_comments                           |
|----------|--------|-----------------------------------------------|
| 1        | 1      | Report is valid and verified for January 2025 |
| 2        | 2      | Report is valid and verified for January 2025 |